

03

LECTURE

HEALTH CARE

MT 1, Second Semester LECTURE 03 - Ms. Marian Helen G. Tecson, RMT

FIRST AID AND PERSONAL WELLNESS

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LEARNING OBJECTIVES

- Define stress and differentiate the 2 forms of stress
- Discuss the physiologic effects, signs, and causes of stress
- Explain the role of stress in maintaining health & wellness
- Identify specific stress management strategies
- Understand personal wellness and the dimensions of wellness
- Define and differentiate the type of shocks
- Learn techniques in first aid: basic life support and CPR

I. UNDERSTANDING STRESS

Stress

- aka **20th Century Syndrome** by **Benjamin Franklin**
- nonspecific response of the body to any demand made upon it
- degree to which you feel overwhelmed or unable to cope
- a feeling of emotional or physical tension
- can come from an event or thought that makes you feel angry, nervous, or frustrated

STRESS VS. ANXIETY

Stress

- caused by existing stress-causing factor or stressors

Anxiety

- Stress continues even when the stressor is already gone
- **Long-term effect of consistent stress**

A. ACUTE VS. CHRONIC STRESS

2 FORMS OF STRESS

Acute Stress

- **most common type**
- **short-term response** to an **imminent danger** or **event**
- helps manage dangerous situations
- Can occur when you do something new or exciting
 - Ex: immediate slam on the breaks as reflex ; skiing a steep slope
 - “Mini heart attack feeling”

Chronic Stress

- **most serious type of stress**
- continuous or **prolonged type**
- more than an individual can cope or control
 - You can become so used to chronic stress that you do not realize it is a problem and may affect health.

Physiologic Effects of Stress

- Body reacts to stress by **releasing hormones**
- **Hormones** make the brain more alert, cause the muscles to tense, increase the pulse
- These reactions are good as they can help you handle the situation causing stress; *the body's way of protecting itself.*

- In **Chronic stress**, body stays alert even in the absence of danger
- Overtime, puts the individual at these risks:
 1. High blood pressure
 2. Heart disease
 3. Diabetes
 4. Obesity
 5. Depression or anxiety
 6. Skin problems (e.g. acne or eczema)
 7. Menstrual problems

B. SIGNS AND CAUSES OF STRESS

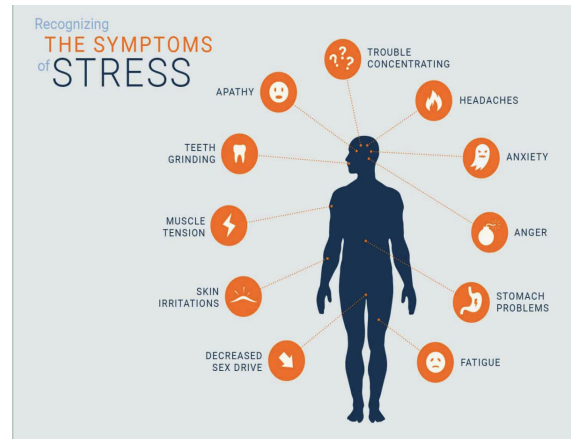
Signs of Stress

- Stress can cause **emotional and physical symptoms**
- There can be emotional, behavioral, and bodily changes
- Sometimes, you may not realize that these symptoms are caused by stress

Symptoms:

- a. Trouble concentrating — loses focus and becomes forgetful
- b. Headaches
- c. Anxiety
- d. Anger
- e. Stomach problems — e.g. diarrhea and constipation
- f. Fatigue — tiredness and lack of energy ; unmotivated all the time
- g. Decreased sex drive
- h. Skin irritations
- i. Muscle tension
- j. Teeth grinding — stiff jaw or stiff neck
- k. Apathy — lack of feeling or emotion — lack of interest or concern
- l. Trouble sleeping or sleeping too much
- m. Weight loss or weight gain

n. Use of alcohol or drugs to relax



Causes of Stress

- Causes are different for each person
- One can have stress from good or bad challenges

Common stressors of college students:

- a. Alcohol use
- b. Drug use
- c. Academic competition
- d. College red tape
- e. Religious conflicts
- f. Choice of major/ future job
- g. Sexual pressure
- h. Family responsibilities
- i. Loneliness, depression, anxiety
- j. Money troubles
- k. Military obligations
- l. Social alienation, anonymity
- m. love/marriage decisions
- n. Illness and injury
- o. Lack of privacy
- p. Parental conflict
- q. Time management

Common stressors of college students



C. EUSTRESS VS. DISTRESS

REACTION TO STRESS THAT AFFECTS WELLNESS

- **Emotional Intelligence (EI), Emotional Quotient (EQ), and Emotional Intelligence Quotient (EIQ)**
 - these three terms have the same definition
 - The capability of individuals to recognize their own emotions and those of others, discern between different feelings and label them appropriately, use emotional information to guide thinking and behavior, and manage and/or adjust emotions to adapt to environments or achieve one's goal(s).
- **Why is Emotional Intelligence so important?**
 - When it comes to happiness and success in life, EQ matters just as much as IQ.

- IQ isn't enough on its own to achieve success in life
- **EQ helps in managing the stress and emotions**
- IQ and EQ exist in tandem and are most effective when they build off one another

Eustress

- **good response to stress**
- refers to happy, pleasant events
- increased performance even if stress increases
- It produces positive feelings of excitement, fulfillment, meaning, satisfaction, and wellbeing.
 - **Dr. Michael Genovese**, a clinical psychiatrist, explained that exciting or stressful events cause a chemical response in the body.
 - He also said that without eustress, our wellbeing can suffer. It makes us feel **motivated, work toward our goals, and feel good about life.**

Distress

- **bad response to stress**
- unpleasant or harmful reaction as the stress increases
- results to unhealthy body and declining performance
 - Licensed professional, counselor **Casey Lee**, says that this type of negative stress can lead to **anxiety, depression, and a decrease in performance**

D. YERKES-DODSON LAW

- A certain amount of anxiety can enhance performance too much can impair it (ie; severe stage fright)
- **Model of the relationship between stress and task performance**
- Also known as **Inverted U Model of Arousal** – depicted as an upside down u-shaped curve
- Was originally developed by psychologists **Robert Yerkes** and **John Dillingham Dodson** in **1908**
 - **Horizontal line (Stress) :**
 - a. Left side = low arousal/ stress
 - b. Middle = medium arousal
 - c. Right side = high arousal
 - **Vertical line (Performance) :**
 - a. Bottom = low performance
 - b. Middle = medium performance
 - c. Upper = high performance



E. GENERAL ADAPTATION SYNDROME (GAS)

- Developed by **Dr. Hans Selye**
- Three stage process that describes the **physiological changes** that the body goes through when it is under stress

THREE-STAGE PROCESS OF STRESS RESPONSE

Alarm Reaction Stage

- **initial symptoms** the body experiences when under stress
- **"fight-or-flight" response**
- increase in heart rate, blood pressure, respiration rate
- blood sugar levels rise, muscles tense up, pupils dilate, and perspiration increases
- The **adrenal gland releases cortisol**, a stress-hormone, and a boost of adrenaline that increases energy

Stage of Resistance

- not as intense as the alarm reaction stage but the **bodily levels of hormonal secretion are still above normal**
- **Autonomic Nervous System (ANS)** tries to return the body to normal by reducing the amount of cortisol produced
- heart rate and blood pressure begin to return to normal
- If the stressor remains, the body will stay in a state of alert and stress hormones continue to be produced
 - **Signs:** irritability, frustration, and poor concentration

Stage of Exhaustion

- **the body has depleted its energy resources** by continually trying but failing to recover from the initial alarm reaction stage
- **result of prolonged or chronic stress**
- Struggling with stress for long periods can drain a person's physical, mental, and emotional resources to the point where the body is no longer equipped to fight stress
 - **Signs:** fatigue, tiredness, burnout, depression, anxiety, decreased stress tolerance, and feeling unable to cope

STRESS MANAGEMENT STRATEGIES



F. SPECIFIC STRESS MANAGEMENT STRATEGIES

1. EXERCISE

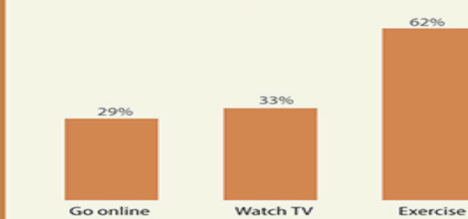
- provides a **natural mood boost** since it stimulates your body to release a number of hormones like **endorphins**
- **ENDORPHINS** – chemicals produced by the body to relieve stress and pain. This also triggers a positive feeling in the body.

Ex.

- Running
- Swimming
- Dancing
- Cycling
- Other forms of physical activity

PEOPLE WHO ENGAGE IN SEDENTARY ACTIVITIES TO MANAGE STRESS ARE LESS LIKELY TO SAY THAT THE TECHNIQUE IS EFFECTIVE COMPARED WITH THOSE WHO ENGAGE IN MORE PHYSICALLY ACTIVE STRESS MANAGEMENT STRATEGIES.

Reported Effectiveness of Stress Management Techniques



BASE: All respondents 2013 (Adults n=1950)
Q965 Do you do any of the following to help manage or relieve your stress?
BASE: Respondents who do activity to relieve stress 2013 (Bases vary)
Q971 How effective is this for helping you manage or relieve your stress?

2. RELAXATION TECHNIQUES

- Meditation
- Autogenic training/ Self-generating imagery
 - is a relaxation technique focusing on **promoting feelings of calm and relaxation in your body** to

help **reduce stress** and **anxiety**

- Progressive muscle relaxation
- Abdominal breathing
- Yoga/Tai Chi
- Massage
- Listening to Relaxing or Soothing Music
- Laughing



3. LIFESTYLE CHANGE:

- Proper nutrition/ diet
 - You should drink 8 glasses of water a day, eat rice, bread, noodles, fruits and vegetables, milk and products, fish, meat, etc. and just a few amounts of sugar or sweets and fats and oil.



b. Improve time management

- **know what to prioritize**, prepare a schedule for each day or have a daily routine and stick to your schedule don't keep adjusting it.

c. Reduce/eliminate alcohol, drugs & cigarette use

d. Get plenty of sleep

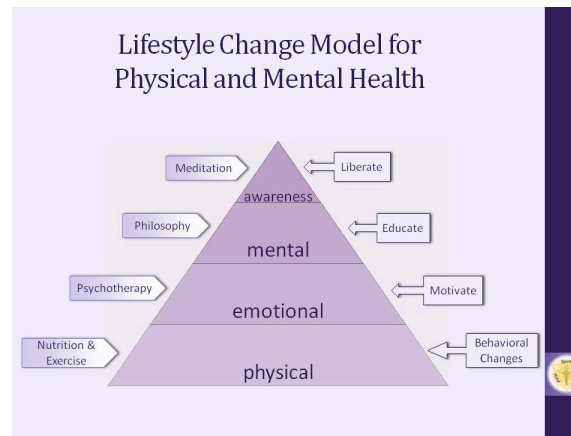
4. REFRAMING:

- Consciously **reinterpreting a situation in a more positive light.**
- Reframe life's stumbling blocks into challenges.
- **Look on the bright side**, take control of your reactions, learn to be an optimist.

OPTIMIST - is a person who tends to be hopeful the confidence about the future

a. Developing satisfying relationships

- b. Learn when to seek help and the support of others
- c. Schedule "me time" and listen to music
- do things that you love to spend time on, doing things that make you happy and don't forget to take care of yourselves. It's not selfish to think of yourselves first from time to time.



5. LAUGHTER AND HUMOR

- Can **provide psychological relief from tension, anxiety, anger, hostility, and emotional pain.**
- Laughing is internal jogging as it causes **endorphins to be released in the brain.** It helps relax the blood vessels and blood circulation
- It provides a **greater sense of control, lowers stress hormones, and improves immune function.**
- Try to see the humor in everyday situations and don't be afraid to laugh at yourself.
- Avoid humor at someone else's expense.

6. CREATE A MEMORY BANK

- Happiness comes from noticing and enjoying the little things in life.

- Savor special experiences of your life and store them in your memory bank (e.g. taking pictures, videos etc.)
- Journaling will help you remember them
- Allow yourself to remember pleasant things and feel happy.

NOTES: FOR STUDENTS!

College is a time of growth and positive life experiences. It can also be a time of stress and anxiety. **Your college experience will depend on how well you manage your stress to reduce anxiety.**

1. Get plenty of sleep

- 7-9 hours
- No all-nighters
- Same schedule every night

2. Eat healthy balanced meals

- Three meals a day
- Small snacks for energy
- Fruits, vegetables and grains

3. Time management

- Create a schedule
- Prioritize assignments and projects
- Reward accomplishments

4. Schedule down time

- Hobbies or exercise
- Time with friends
- Make it a part of your schedule

NOTES: IN THE WORKPLACE

Ergonomics

- Greek derivation
 - *Ergo* (work)
 - *Nomos* (law)
- Interfacing man, machine, and environment to optimize safety, productivity, and comfort.

"Fitting the task to the human."

VIDEO: Importance of Ergonomics in the Workplace

<https://www.youtube.com/watch?v=GuhhmNjJVHE&t=5s>

Ergonomics & Posture

- Lessen the pain we may experience on our desk

We can reduce risks with these ergonomic principles:

- Make sure to have good posture
- Use large muscle groups when possible
- Bend from the middle to extend your range of movement

Musculoskeletal Disorders (MSD)

- Occur when the spine joints, nerves and muscles are injured.

Work Related MSDs (MSD)

- Occur when the work environment or activities aggravate the condition
 - Eye Strain
 - Elbow Pain
 - Lower Back Pain
 - Carpal Tunnel Syndrome
 - Rotator Cuff Injury
 - Tendonitis
 - Sprains, Strains, & Tears
- **MSD's affect the::**
 - Muscles
 - Ligaments
 - Tendons
- **MSD's increase the risk of:**
 - Neck injuries
 - Nerve Damage
 - Back Problems

Impact of MSD's In the Workplace

- In 2014, the **Bureau of Labor Statistics** found that occupations with the highest WMSD included:
 - Construction
 - Healthcare
 - Retail & Wholesale Trade
 - Transportation
 - Warehousing

Preventing MSDs in the Workplace

Employee Programs

- **Healthy Behavior Programs**
 - Nutrition
 - Tobacco Use
 - Physical Activity
 - Alcohol & Substance Abuse
 - Mental Health
 - Depression Screening
 - Support Programs
- **Health Screening**
 - Blood Pressure
 - Obesity
 - Cancer
 - Diabetes
 - Cholesterol
 - Immunizations
 - Flu
 - Pneumonia

Steps to Prevent Workers From MSDs

1. Provide Management Support

- Management is responsible for setting goals and objectives for the economic plan

2. Involve Everyone

- Allowing everyone to participate provides valuable information about the risks in the workplace

3. Provide Training

- Having proper training ensures proper education of the benefits of ergonomics and the value of reporting symptoms of WMSD

4. Identify Problems

- it is important to recognize and evaluate economic issues in the workplace before they result in WMSD

5. Encourage Reporting of WMSD Symptoms

- reporting symptoms early can accelerate the improvement method and help reduce or prevent the development of serious injuries the progression of symptoms and subsequent loss work time

6. Implement Solutions

- multiple solutions exist for reducing controlling or eliminating workplace MSD's

7. Evaluate Progress

- as an economic process develops assessment should involve deciding if the goals have been met to help determine the success of the economic solutions

VIDEO: How to Set Up Your Desk Ergonomically

https://www.youtube.com/watch?v=F8_ME4VwTiw

STEP 1: Adjust Your Chair

- Make sure your elbows are bent to **90 degrees**
- If a person's feet are not touching the floor, give a footstool. (If you don't have a foot rest, use a rim of paper.)

STEP 2: Adjust Your Monitor

- Arm Length
- Top of the screen is eye level
- If your monitor is not adjustable in height use your reams of paper

STEP 3: Mind Your Mouse & Keyboard

- Where your hands end up is where your keyboard should be
- Your mouse should end up right next to your keyboard

STEP 4: Position Your Phone

- Put phone on your non-writing side
- Use a headset if you are on the phone a lot during the day.

STEP 5: Move (& Stretch Out)

- move after 10-15 minutes

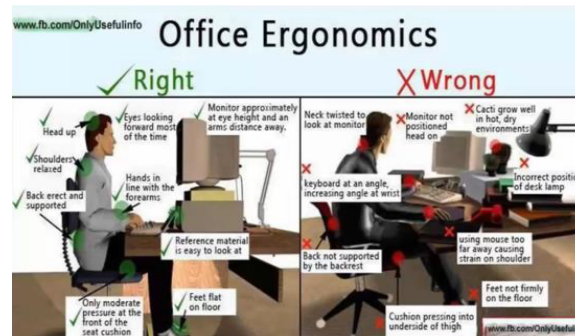
Basic Exercises You Can Do While Sitting in Your Chair

- First exercise is a chin tuck
- Second exercise is for your upper traps you're going to do a basic stretch where you bend your head to one side and then gently pull for a little more
- Third exercise is called the scapular retraction you're basically going to squeeze your shoulders back
- Fourth exercise is for your lower back this is what we call a pelvic tilt

The most important thing you want to do is;

- Get up out of your chair every hour and walk
- Get something to eat or something to drink

A. OFFICE ERGONOMICS

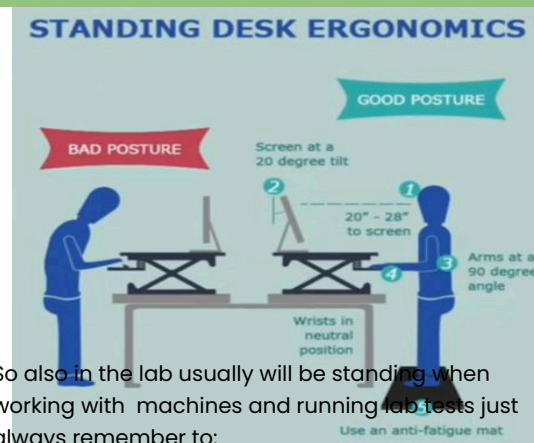


Correct Things to do in Office Ergonomics

- The monitor is placed **approximately at eye level** and at an **arm's distance away**
- The person is sitting on the chair with his **head up** and **eyes looking forward**
- Shoulders relaxed, back erect and supported
- His hands are in line with his four arms
- Feet flat on the floor he/she is able to sit up straight

These are really applicable when you have your online class. The right side of the photo shows everything that is wrong, so you should avoid these errors.

B. LABORATORY ERGONOMICS



So also in the lab usually will be standing when working with machines and running lab tests just always remember to:

- Maintain good and proper posture to minimize the risk of injury or harm.

II. PERSONAL WELLNESS

- The **World Health Organization (WHO)** defines personal wellness:

◦ "As the state of complete physical, mental, and social well-being, and not merely the absence of disease or infirmity."

- We actually use this to define health in PMLS ethics. That's them but this also still relates to personal wellness.

- The **National Wellness Institute (NWI)** defines personal wellness:

◦ "A conscious, self-directed, and evolving process of achieving full potential."

- Once you're able to manage your stress then you can most likely achieve personal wellness.

A. DIMENSIONS OF WELLNESS



There are eight dimensions of wellness namely:

1. Intellectual
2. Emotional
3. Physical
4. Social

5. Occupational
6. Financial
7. Environmental
8. Spiritual

Emotional

- Coping effectively with life and creating satisfying relationships

Financial

- Satisfaction with current and future financial situations.

Social

- Developing a sense of connection, belonging, and a well-developed support system.

Spiritual

- Expanding our sense of purpose and meaning in life.

Occupational

- Personal satisfaction and enrichment derived from one's work.

Physical

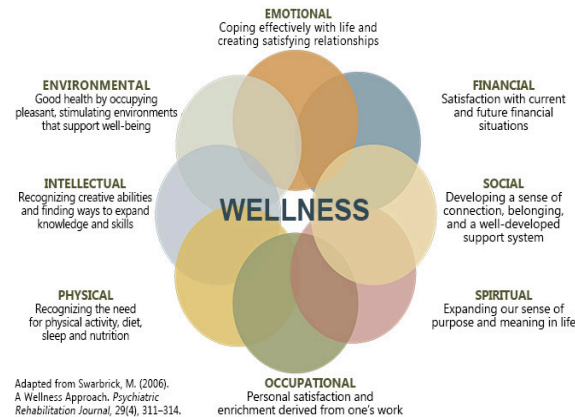
- Recognizing the need for physical activity, diet, sleep and nutrition.

Intellectual

- Recognizing creative abilities and finding ways to expand knowledge and skills.

Environmental

- Good health by occupying pleasant, stimulating environments that support well-being



B. EFFECTS OF STRESS; SHOCK: TYPES, SIGNS & SYMPTOMS

STRESS AS AN EFFECT OF SHOCK

SHOCK	PERFUSION
Is a life-threatening situation due to poor tissue perfusion with impaired cellular metabolism , manifested in turn by serious pathophysiological abnormalities . - Bailey and love	• Is the passage up fluid through the circulatory system to an organ or tissue • usually referring to the delivery of blood to a capillary bed and tissues.

is a term used to describe the **clinical syndrome** that develops when there is **critical impairment of tissue perfusion due to some form of acute circulatory failure**. - Davidson's

- May be simply defined as **inadequate delivery of oxygen and nutrients to maintain tissue and cellular function**
- This is the state in which profound and widespread reduction of effective tissue perfusion leads to reversible, if prolonged, to irreversible cellular injury.

Defined as **inadequate delivery of oxygen and nutrients to maintain normal tissue and cellular function**. - Schwartz's

The state in which **profound and widespread reduction of effective tissue perfusion leads first to reversible**, and then if **prolonged, to irreversible cellular injury**. - Kumar and Parrillo, 1995

Types of Shock

CLASSIFICATION	ETIOLOGY	UNDERLYING PATHOLOGY
HYPOVOLEMIC	Hemorrhage, burns, excessive diuretic use, fluid losses (vomiting, diarrhea)	Whole blood loss, Plasma loss
CARDIOGENIC	MI, dysrhythmia, blunt cardiac injury, valvular disease, and stage cardiomyopathies	Loss of cardiac contractility, reduced CO
NEUROGENIC	Spinal cord injury, vasomotor center depression (drugs, emotional stress)	Decrease in venous return, Poor distribution of blood
ANAPHYLACTIC	Drugs, insect bites/stings, contrast media, blood transfusions, food, venom (bees, etc.)	Decrease in venous return, Poor distribution of blood
SEPTIC	Infection, patients receiving immunosuppressive therapy,	Decrease in venous return, Poor distribution of blood

	malnourished, AIDS, cancer	
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**Etiology- means the cause/s of a certain type of shock.*

Hypovolemic

- **Hypo** - low or decreased.
- **Volemic** - Volume of blood in the body.
 - Cause: Hemorrhage, burns, excessive diuretic use, fluid losses, (vomiting, diarrhea)
 - Underlying Pathology: Whole blood loss, Plasma loss

Cardiogenic

- **Cardio** - heart
- **Genic** - produced/caused by
 - Cause: MI, dysrhythmia, blunt cardiac injury, valvular disease, end stage cardiomyopathies
 - Underlying Pathology: Loss of cardiac contractility, reduced CO

Neurogenic

- **Neuro** - relating to the never or the nervous system (this includes the spine)
- **Genic** - produced/caused by
- Poor distribution of blood
 - Cause: Spinal cord injury, vasomotor center depression (drugs, emotional stress)
 - Underlying Pathology: Decrease in venous return

Anaphylactic

- Derived by the Greek words:
 - **Ana**- against/back
 - **Phylaxis**- protect/guard

- **Anaphylactic Shock** - caused by **severe allergic reactions**.
 - Cause: Drugs, insect bites/stings, contrast media, blood transfusions, food, venom (bees, etc.)
 - Underlying Pathology: Decrease in venous return, Poor distribution of blood

**Contrast media - it is used in imaging techniques to enhance the differences between body tissues on images*

Septic

- Derived from Greek Words
 - **Sēptikos**: putrid or rotten
 - **Sēpein**: to make it rotten
 - **Suffix -ic**: - of or pertaining to
- Septic Shock - from the word itself, is **from infections**.
 - Cause: Infection, patients receiving immunosuppressive therapy,
 - malnourished, AIDS, cancer.
 - Underlying Pathology: Decrease in venous return, Poor distribution of blood

Signs and Symptoms of Shock

- Cool, clammy skin
- Pale or ashen skin
- Rapid pulse
- Rapid breathing
- Nausea or vomiting
- Enlarged pupils
- Weakness or fatigue
- Dizziness or fainting
- Changes in mental status or behavior, such as anxiousness or agitation

III. FIRST AID

- If you suspect a person is in shock, call 161 (ERUF) or your local emergency number. Then immediately take the following steps:
 - **Lay the person down** and **elevate the legs and feet slightly**, unless you think this may cause pain or further injury.
 - **Keep the person still** and don't move him or her unless necessary
 - **Begin CPR if the person shows no sign of life**, such as breathing, coughing, or movement.
 - **Loosen tight clothing** and if needed, cover the person with a blanket to prevent chilling
 - **Don't let the person eat or drink anything**
 - If you suspect that the person is having an allergic reaction, and you have access to an epinephrine auto-injector, use it according to its instructions.
 - If the person is bleeding, **hold pressure over the bleeding area**, using a towel or sheet.
 - If the person **vomits or behind bleeding from the mouth**, you **turn him or her onto a side to prevent choking unless you suspect spinal injury**.

A. BASIC LIFE SUPPORT

- Involves essential emergency procedures to maintain breathing and circulation in a person experiencing cardiac arrest or respiratory failure.
- Includes CPR (chest compressions and rescue breaths), using an automated external defibrillator (AED), and clearing airway obstructions

- **Aims to sustain life until professional medical help arrives, improving survival chances.**

In a life and death situation, actions must be taken to ensure that lives can be saved.

3 IMPORTANT STEPS!

Step 1: Secure the scene

- This means **pulling the victim out of dangerous situations**, e.g. standing water, traffic, and other dangerous situations.
- **ENSURE THAT YOU, YOURSELF ARE SAFE BEFORE PERFORMING BASIC LIFE SUPPORT**

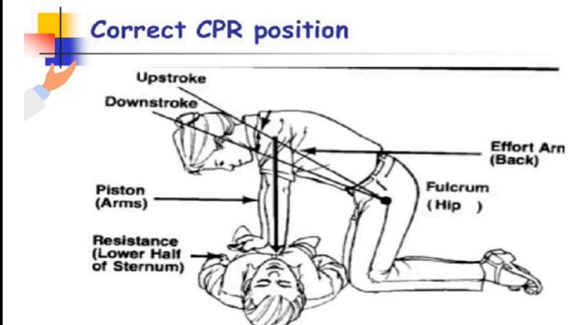
Step 2: Check for a response

- Shake shoulders gently
- Ask "are you all right?" or other types of questions that may induce a response
- If they **respond**:
 - Leave as you find them (DON'T touch them, not leave as if biyaan)
 - Find out what is wrong
 - Reassess regularly
- IF the victim is **UNRESPONSIVE**, and has abnormal respirations
 - *no respiration or gasping/agonal (Shallow, slow, and infrequent gasps) breathing*
 - activate EMS (emergency medical services)
 - yell for help
 - send someone for an AED (automatic external defibrillator)
 - use a cell phone if one is available

Step 3: Check for a pulse

- Check for a pulse at the **carotid artery and check breathing**
- *Check no longer than 10 second*
- If there is a pulse and breathing, continue to monitor and support breathing
- if there is a pulse and no/abnormal breathing, start rescue breathing

***If there is no pulse and no/abnormal breathing: Start High Quality CPR**



B. CPR

- A **life-saving technique** used in emergencies when a person's breathing or heartbeat has stopped.
- Involves **chest compressions** (pushing hard and fast in the center of the chest) and **rescue breaths** to **restore circulation and oxygen flow**.
- Increases survival chances by maintaining blood flow to vital organs until professional medical help arrives.

High Quality CPR

- 30 compressions to 2 breaths
- 100-120 compressions per minute

IMPORTANT!

1. Compress the chest between **2 and 2.4 inches**
2. Allow the chest to **FULLY recoil**
3. Continue CPR for 2 minutes OR until AED is on the victim, powered up, and ready for use. Check heart rhythm using AED.
4. Follow instructions on AED., deliver shock as needed

ADDITIONAL INFORMATION!!!

Automatic External Defibrillator

- a **portable, life-saving device** used to treat **sudden cardiac arrest** by analyzing the heart's rhythm and delivering an electrical shock

- Once turned on, the device guides the user with voice prompts to place electrode pads on the victim's bare chest. The machine automatically analyzes the heart rhythm and instructs the user to press a button to deliver a shock if necessary

EMS FILIPINO NUMBERS

- Nationwide Emergency Hotline:
 - 911
- Philippine Red Cross:
 - 143

CPR-ADULT AND CHILD

CPR (Cardiopulmonary Resuscitation)

- Emergency procedure used when breathing or heartbeat has stopped.

Purpose:

- Maintain blood circulation
- Provide oxygen to the brain and vital organs
- Prevent brain damage
- Brain damage can begin within 4-6 minutes without oxygen.

When to Perform CPR?

Person is:

- Unresponsive
- Not breathing or only gasping
- Has no pulse (for trained responders)

Initial Response Steps

1. Check for Responsiveness

- Tap shoulders and shout: "Are you okay?"
- If no response → proceed immediately.

2. Call for Help

- Activate emergency response system.
- Call the local emergency number.
- Ask someone to get an AED (Automated External Defibrillator) if available

3. Check Breathing

- Look for normal breathing (5-10 seconds).
- Occasional gasps are NOT normal breathing.

ADULT CPR

1. Handplacement

- Place the heel of one hand on:
 - Center of the chest
 - Lower half of the sternum
- Place second hand on top:
 - Interlocked fingers
 - Keep arms straight
 - Shoulders directly above hands

2. Chest Compressions

- Depth: At least 2 inches (5cm)
- Rate: 100-120 compressions per minute
- Allow full chest to recoil
- Minimize interruptions

3. Compression-To- Breath Ratio

- 30 compressions: 2 breaths
- Continue Cycle until:
 - AED arrives
 - Professional help arrives
 - Person shows signs of life
 - You are too exhausted to continue

4. Rescue Breaths

- Open airway using:
 - Head-tilt, chin-lift
 - Pinch nose closed.
 - Give 1 breath over 1 second.
 - Chest should visibly rise.
 - Give second breath.
 - Avoid excessive ventilation.

CHILD CPR (1 YEAR TO PUBERTY)

DIFFERENCES FROM ADULT CPR

1. Handplacement

- Use:
 - One hand (for smaller child)
 - Two hands (if needed for depth)
- Same Location: Center of Chest

2. Compression Depth

- About – About 2 inches (5 cm)
- Or about 1/3 depth of chest

3. Compression Rate

- Same as adult: 100–120 per minute

4. Compression to Breath Ratio

- Single rescuer: 30:2
- Two Rescuers: 15:2

- Push hard and push fast.
- Do not delay compressions.
- Allow full chest recoil.
- Avoid long pauses.
- Continue CPR until help arrives.

IV. REFERENCES

- A Wellness Way of Life by Gwen Robbins
- Achieving Your Wellness Workbook by V. Thondley
- Nursing Care of the Community by Z. Famorca
- www.americanredcross.org
- www.health.harvard.edu

Use of AED (Automated External Defibrillator)

Steps:

1. Turn on AED.
2. Attach pads to bare chest.
3. Follow voice prompts.
4. Ensure no one is touching the patient during analysis.
5. Deliver shock if advised.
6. Immediately resume CPR after shock

ADDITIONAL PEDIATRIC AED NOTES

- If not available, use adult pads (ensure they do not touch).
- Follow instructions on AED., deliver shock as needed

IMPORTANT REMINDERS!!!